

23. REVISION D1: NOTOCHORDAL PROCESS

DURATION : 03:09

STUDY SHEET

COURSE SUMMARY

In this session, we explore the notochordal process, which is essential for embryonic formation. We study the dome structure, where the ectodermal tissue loses its polarity and becomes a fulcrum for development. The aspiration field generated by the primitive streak draws epithelial cells towards the centre, leading to the formation of the mesoderm. The mechanisms involved, such as the release of hyaluronic acid and the dynamics of cell growth, are examined in detail, highlighting the crucial importance of these transformations in early embryonic development.

DAY 1 REVISION: NOTOCHORDAL PROCESS

We will examine the **notochordal process** to deepen your understanding.

Imagine a **dome-shaped** structure. The **embryonic pedicle** is not present here, and therefore the trophic source is located at this point.

The **ectodermal tissue** above is no longer polarised and no longer absorbs. This becomes a fulcrum. It is important to visualise that everything is **growing**. Everything organises around and creates itself.

This becomes **Hensen's node**. Simultaneously, everything acts as a **suction field**. A space will open between the ectoderm and the endoderm, between the epiblast and the endoderm.

This space generates, at the level of the **primitive streak**, this suction field. The cells, which are **epithelial** cells, are drawn towards the centre.

They are pulled towards the centre, compress, become **bottle cells**, pass underneath, and release an acid, **hyaluronic acid**. This acid will be very important later in protein and structural formations, but that is not the main point for now.

These cells pass underneath and fill the space. This is how the **mesoderm** develops.

We have this wave, this suction field that creates a primitive streak. This movement pulls the cells towards the centre. Meanwhile, Hensen's node progresses.

I am always at the shortest point. I do not move. What am I? The **zero point**. And I observe all of this.

In parallel, in this same development, there is **nodal fluid** inside. Since it is in the notochord, it is called nodal fluid.

Why do the cells in the dome develop faster? It has been observed that cells in this expanded process, as if we were no longer in an expansion dome, multiply much faster than those on the same line, but which are converging.

These latter develop less quickly.